



University
of Glasgow

Lecture 5-Part 1: Qualitative and Quantitative Research Methods

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Today's session

- Summarise the videos
- Quantitative research methods
- Qualitative research methods
- Data collection: the 5 Ps
- A note about some data collection terminology
- Data collection methods
- Analysing qualitative data: open, axial, selective coding

Video: Sample Sizes & Statistical Powers

Video — Sample Sizes & Statistical Powers

Spoke in the past about how it isn't feasible to evaluate an entire population in an experiment:

- Instead observed a subset of a population and draw conclusions for a wider population

Note: We need to be careful about this sample size — they must have the characteristics of the wider population being studied

Video — Sample Sizes & Statistical Powers

Video also covers statistical powers — power analysis when possible

Also the number of participants used in an experiment:

- **Discusses “local standards” in HCI as the example**
- **Aim to do (roughly) what peers in the same domain are doing**

Video: Data Analysis

Video — Data Analysis — data collection

We'll speak a bit more about this today as well:

- **This gives you an overview of how you go from “__ is the variable to measure” to “collecting data to investigate the variable __”**
- **Need to ensure that the collected data sufficiently answers a hypothesis, and can be easily analysed to answer the hypothesis/RQs**

Video — Data Analysis — statistical analysis

Discusses a number of statistics we can use to test a hypothesis:

- **Focuses on null-hypothesis significance testing**
- **Introduces some of the statistical measures you might use to do this**

Need to be aware how we use statistics to draw a conclusion about a hypothesis

Video — Data Analysis — ethics

- The video highlights a number of the ethical frameworks researchers must abide by when conducting experimental work (particularly important if your experimental work involves human participants)
- Goes over the ethical process here at Glasgow to give you a sense for how this works here — and goes over key ethical points which must be adhered to during experimental work

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Quantitative research methods

- Quantitative methods discussed in the video.
- A method of research that relies on measuring variables using a numerical system, analyzing these measurements using any of a variety of statistical models, and reporting relationships and associations among the studied variables.
- Quantitative research data is often standardized and can be easily used to generalize findings

Characteristics of Quantitative research methods

1. Numerical Data:

Quantitative research focuses on quantifiable data, which can be measured and expressed numerically. Examples include survey results, statistical data, and experimental measurements.

2. Objectivity and Rigor:

The research aims to remain objective and free from researcher bias. This is achieved through standardized methods, clear protocols, and statistical tools.

Characteristics of Quantitative research methods

3. Structured Methodology:

Quantitative research follows a structured process, including the formulation of hypotheses, data collection, analysis, and conclusion.

4. Use of Statistical Analysis:

Data is analyzed using statistical techniques to identify patterns, relationships, and causal effects. Descriptive and inferential statistics are commonly applied.

Characteristics of Quantitative research methods

5. Large Sample Sizes:

Quantitative studies often involve larger sample sizes to ensure the reliability and generalizability of the results.

6. Hypothesis Testing:

Researchers often begin with a specific hypothesis or research question and test it using empirical data.

Characteristics of Quantitative research methods

7. Generalizability:

The goal is to produce results that can be generalized to a larger population, provided the sample is representative.

8. Controlled Variables:

Experiments and studies are designed to control variables and minimize confounding factors to isolate the effects of the variable being studied.

Characteristics of Quantitative research methods

9. Reproducibility:

The methods are documented in detail, allowing other researchers to replicate the study to verify results.

10. Deductive Approach:

Quantitative research often uses a deductive reasoning process, moving from theory or hypothesis to data collection and analysis to confirm or refute the theory.

Characteristics of Quantitative research methods

11. Focus on Relationships and Patterns:

Quantitative research aims to identify trends, relationships, or causation between variables.

12. Use of numerical tools and Instruments:

It employs tools like surveys, questionnaires, structured interviews, and experiments to collect data systematically.

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What is Qualitative Research?

A method of inquiry focused on understanding **human behaviour** and the reasons that govern such behaviour. It emphasizes **subjective assessments** and the exploration of phenomena in their natural settings through methods like interviews, observations, and analysis of texts and artifacts.

Early development of qualitative research is often associated with the field of sociology and anthropology.

Qualitative Research in Computer Science

Although most of scientific research is quantitative, qualitative research methods are common in some areas, such as: HCI, CS education and Software Engineering.

Examples:

- SE: Investigating software adoption or user behaviour towards a new release
- HCI: How do users perceive and experience the integration of artificial intelligence in everyday personal technology devices?

Combining quantitative and qualitative methods are referred to as “mixed methods”: can provide a comprehensive understanding of a research problem.

Qualitative Research Characteristics

1. **Data Collection:** Qualitative research often involves collecting data through direct observation, interviews, questionnaires, focus groups, participants' diaries, or other methods that allow researchers to access the perspectives and experiences of their subjects.
2. **Subjective Perspective:** This approach often seeks to understand how individuals construct their own meaning and experience of the world. It acknowledges that the researcher's own background and perspective can influence the interpretation of the data.

Qualitative Research Characteristics

3. **Exploratory Nature:** It is typically exploratory and is used to gain an understanding of underlying reasons, opinions, and motivations. It provides insights into the problem or helps to develop ideas or hypotheses for potential quantitative research.
4. **Contextual Detail:** Qualitative research is good at providing rich, detailed descriptions that bring out the nuances of complex situations or phenomena.

Qualitative Research Characteristics

5. **Analysis:** Data analysis in qualitative research is often more interpretative, subjective, and involves identifying patterns and themes in data.
6. **Flexibility:** Research design in qualitative research is often less structured than in quantitative research, and can evolve as the research progresses.
7. **Inductive Approach:** In qualitative research, the focus is on exploring, understanding, and interpreting the meanings and experiences of participants. Researchers observe, gather rich, detailed data (like interview transcripts, observations, or documents), and then derive patterns, themes, and theories from this data.

Qualitative vs. Quantitative Research

	Quantitative Research	Qualitative Research
Objective	To quantify data and generalize results	To understand meanings, concepts, characteristics
Data Type	Numeric, measurable	Textual, observational
Approach	Deductive (testing theories)	Inductive (developing theories)
Focus	Broad, generalizable	Deep, detailed
Collection Methods	Surveys, structured interviews, measurements	Interviews, focus groups, observations, textual analysis
Analysis	Statistical, mathematical	Thematic, content analysis, narrative
Outcome	General laws, predictive models	Descriptive, interpretative
Sample Size	Large for statistical significance	Smaller, more focused
Flexibility	Structured and rigid	Flexible, adaptable
Researcher's Role	Objective observer	Often participative, subjective
Results	Often in charts, graphs	Often in words, images, themes

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Qualitative Data Collection Methods

Most common: interviews, focus groups, observations, case studies, ethnographic studies, document analysis, artifact analysis and participatory design.

When to use each method?

Depends on several factors: research objectives and questions (Specificity vs. Exploration), depth of Information needed, topic's sensitivity and complexity, participants' accessibility and willingness, resources available (time, budget and expertise), data analysis plans and ethical considerations.

Interviews

Conducting **one-on-one** or **group** interviews

Types of interviews: structured, semi-structured, and unstructured.

Example: Interviewing users to understand their experiences with privacy settings on social media platforms.

Focus Groups

Facilitating discussions with a group of users to gain multiple perspectives on a topic.

Example: Holding a focus group to discuss the usability of a new mobile application.

Observations

Observing users in their natural environment as they interact with technology.

Practical considerations and challenges.

Example: Watching how children interact with educational software in a classroom setting to identify intuitive design elements.

Case Studies

In-depth examination of a single instance, event, individual, or group to explore and understand complex issues in real-life contexts.

Utilizes multiple sources like interviews, documents, reports, observations, and sometimes physical artifacts.

Example: A case study of the implementation and adoption of a new healthcare information system in a hospital.

Ethnographic Studies

Immersing in the environment and everyday life of the subjects to understand the cultural context.

Long term, to understand the culture, practices, and experiences of the group from an insider's perspective.

Example: Living in a tech start-up community to study the culture of innovation and its impact on software development practices.

Document Analysis

Analysing documents, reports, e-mails, and other forms of written communication related to the use and development of technology.

Example: Reviewing GitHub commit messages and issue discussions to study collaboration patterns among developers.

Qualitative Data Sampling Methods

Data sampling refers to the process of **selecting individuals, groups, events, or instances to study**. Unlike quantitative research, where the aim is often to select a sample representative of a larger population, qualitative sampling focuses on depth and richness of data.

Common qualitative sampling methods: purposeful sampling, snowball sampling, convenience sampling, criterion sampling, homogeneous sampling, maximum variation sampling

Qualitative Data Sampling Methods

- Purposeful Sampling:

- Selecting individuals or cases that have specific characteristics or experiences relevant to the research question.
- *Example:* Studying the adoption of a new programming language by selectively interviewing software developers who have switched to that language in the past year.

- Snowball Sampling:

- Current study participants recruit future participants from among their acquaintances. Useful when studying hidden or hard-to-reach populations.
- *Example:* Researching cybersecurity practices among a specific group of IT professionals where initial contacts help identify other participants.

Qualitative Data Sampling Methods

- Convenience Sampling:

- Choosing individuals who are easily accessible. While convenient, this method may not provide a diverse cross-section of the population.
- *Example:* Interviewing computer science students available on a university campus to understand their learning challenges.

- Criterion Sampling:

- Selecting cases that meet some predetermined criterion of importance.
- *Example:* Investigating user experience design by sampling users who specifically use adaptive and assistive technologies.

Qualitative Data Sampling Methods

- **Homogeneous Sampling:**

- Selecting a small, homogeneous group to describe or understand a particular subgroup in depth.
- *Example:* Focusing on a specific group of users, like teenage gamers, to explore their interaction with gaming interfaces.

- **Maximum Variation Sampling:**

- Capturing a wide range of perspectives by selecting participants who represent a broad variety of experiences related to the research question.
- *Example:* Exploring the impact of remote work on software development by interviewing a diverse range of software developers from different companies, roles, and geographical locations.

Ethical Considerations

Confidentiality and Consent: Ensure your method allows for ethical considerations like informed consent and confidentiality.

Impact on Participants: Consider how the data collection process might affect the participants, especially in sensitive topics.

Bias:

- Handling bias is crucial to maintain the integrity and credibility of the findings.
- Qualitative research, by its nature, involves subjective interpretation
- Methods to manage and mitigate bias: acknowledgement, transparent documentation, triangulation, purposeful sampling.

Qualitative Data Analysis

- Sometimes coding schemes are used. These codes can be derived inductively (from the data itself) or deductively (based on your research questions or theoretical framework).
- Identifying Themes, Patterns, and Relationships
- Reflecting on biases and assumptions

Writing Qualitative Research Reports

How to present qualitative research findings?

Structuring a qualitative research report: reporting on qualitative research involves a detailed and systematic presentation of the methods, findings, and interpretations.

Writing Qualitative Research Reports

- Good Methodology Description:
 - Describe **the research design** (e.g., case study, ethnography).
 - Explain the selection of **qualitative methods** (e.g., interviews, focus groups, observations).
 - Detail the **participant selection** process (sampling method, criteria, demographics).
 - Discuss the **data collection process** (how, when, where data was collected).
 - Explain the procedures for **data analysis** (coding, thematic analysis, narrative analysis).
 - Address **ethical considerations** (consent, confidentiality, data handling).

Writing Qualitative Research Reports

- Good Results / Findings reporting:
 - Present the data in a narrative format, often structured around themes, patterns, or categories identified in the analysis.
 - Use quotes from participants to illustrate points, ensuring they are representative and contextually appropriate.
 - Be transparent about the volume of data and the range of perspectives or experiences captured.

Mixed-Methods Research

Combines **quantitative and qualitative research methodologies** to offer a more comprehensive understanding of research questions, common in many Computer Science fields.

Mixed-Methods Research

- **Combining Methodologies:** It integrates quantitative methods (like experiments, surveys, and statistical analysis) with qualitative methods (like interviews, focus groups, and thematic analysis).
- **Sequential Approaches:** e.g. start with a qualitative exploratory study to further develop the research question and hypothesis then follow with a quantitative evaluation approach

Mixed-Methods Research Examples in CS

- **User Experience Research:** Understanding how users interact with software or systems. Quantitative data might include usage statistics, while qualitative data could come from user interviews or focus groups.
- **Software Development:** Quantitative methods might assess system performance, while qualitative methods could explore user satisfaction or usability issues.

Example research paper

- Nastjuk, Ilja, et al. "What drives the acceptance of autonomous driving? An investigation of acceptance factors from an end-user's perspective." *Technological Forecasting and Social Change* 161 (2020): 120319.

Abstract—Autonomous driving is believed to provide numerous benefits for individuals and society, including increased road safety, reduced traffic congestion, and an improved ecological footprint. However, many barriers still hinder the widespread acceptance of autonomous vehicles. Research has proposed governmental policy strategies to accelerate the diffusion of autonomous driving, but less is known about end-user perceptions of this innovative technology. First, we employ a **qualitative research design to identify the elements attributed to individual acceptance of autonomous driving**. Furthermore, we organize a research model based on the technology acceptance model, validated with an online survey of 316 participants. The findings reveal how social influence, system characteristics, and individual factors determine individual acceptance of autonomous driving. The research helps to strengthen the existing body of knowledge by highlighting individual perceptions, with implications for practitioners.

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10- minutes break